

Sudden cardiac death in athletes

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A 'paradox of sport' is that in addition to the undisputed health benefits of physical activity, vigorous exertion may transiently increase the risk of acute cardiac events. In general, the risk of sudden cardiac death (SCD) approximately doubles during physical activity and is 2- to 3-fold higher in athletes compared to nonathletes. The incidence of SCD in young athletes is in fact very low, at around 1–3 per 100 000, but attracts much public attention. Variations in incidence figures may be explained by the methodology used for data collection and more importantly by differences between subpopulations of athletes. The incidence of SCD in older (≥ 35 years) athletes is higher and may be expected to rise, as more and older individuals take part in organized sports. SCD is often the first clinical manifestation of a potentially fatal underlying cardiovascular disorder and usually occurs in previously asymptomatic athletes. In the young (< 35 years), SCD is mainly due to congenital/inherited cardiac abnormalities, whilst coronary artery disease

(CAD) is the most common cause in older athletes. Cardiac screening including family/personal history, physical examination and resting electrocardiogram (ECG) may identify individuals at risk and has the potential to decrease the risk of SCD in young athletes. Screening including the ECG has a high sensitivity for underlying disease in young athletes, but the specificity needs to be improved, whereas the sensitivity of screening without the use of ECG is very low. The screening modality recommended for young athletes is of limited value in older athletes, who should receive individualized screening with cardiac stress testing for patients with high risk of underlying CAD. As cardiovascular screening will never be able to identify all athletes at risk, adequate preparedness is vital in case of a potentially fatal event at the sporting arena/facility. Firstly, we will review the magnitude of the problem of SCD in athletes of different ages, as well as the aetiology. Secondly, we will focus on how to prevent SCD in athletes of all ages, reviewing cardiovascular screening recommendations as well as emergency preparedness and arena safety.

Keywords: arena safety, athletes, cardiac screening, sudden cardiac death.

Introduction

The sudden death of a young athlete is a rare event, but each case is a tragedy and receives much public attention. The possibility that a young, healthy athlete may collapse and even die whilst physically active is difficult to comprehend; in particular because the most common cause of sudden death in young athletes is sudden cardiac death (SCD), that is, it is the heart (the 'symbol of life') that is malfunctioning. In the majority of SCD cases, there is an underlying cardiac disorder [1], which is often unknown prior to the event. Because of the perceived low incidence of SCD during physical activity, the relevance of the problem of SCD in athletes is debated.

However, in older (≥ 35 years) athletes, the problem is more common and likely to increase. On the one hand, regular physical exercise is associated with a magnitude of positive effects on health [2, 3]. For example, there is strong evidence for an association between physical exercise and a decrease in all-cause mortality [3], as well as a decrease in metabolic disorders (such as dyslipidaemia and diabetes mellitus) and cancers [4, 5]. Habitual aerobic physical activity may reduce the risk of acute myocardial infarction and SCD [6, 7]. On the other hand, the report by Paffenbarger *et al.* [8] of sudden fatal cardiac events in athletes was of great concern not only within the sports community. The 'paradox of sport' is that, in addition to the undisputed health benefits of physical activity, vigorous

exertion may transiently increase the risk of acute cardiac events [6, 9]. It has been demonstrated that the risk of SCD approximately doubles during physical activity (until about 1 h after its cessation) [6, 10].

Ultimately, it is clear that the multitude of health benefits due to regular physical exercise far outweigh the acutely increased risk of SCD [6]. However, the overall benefit/risk ratio would be considerably improved if susceptible individuals with an underlying cardiac condition leading to a higher risk of SCD could be detected early [11, 12]. For this purpose, systematic preparticipation screening of subjects involved in sporting activities has been introduced to try to identify individuals at increased risk (with an underlying disorder), ideally before the onset of symptoms, with the aim of reducing their risk of sudden death (SCD) [13, 14].

In this review, we will focus first on the rate and aetiology of SCD in athletes of different ages. Then, we will consider cardiovascular screening recommendations as well as emergency preparedness and arena safety for the prevention of potentially fatal events [including sudden cardiac arrest (SCA)].

Incidence of sudden cardiac death

An annual incidence of SCD of about 0.7–3.0 per 100 000 athletes is an established estimate in young individuals (age <35 years) [1, 9, 11, 15–17], although a wide range of incidences has been reported (Table 1). For example, a recent study in North American half marathon and marathon runners demonstrated a very low incidence (0.5 per 100 000), of fatal events [18], whereas data from North American military recruits suggested an incidence of SCD in a population of young, competitive, physically active individuals almost

five times higher than the established estimate [19].

These variations may be explained by the methodology used for data collection of fatal events (e.g. media reports or databases) and, more importantly, by differences between subpopulations of athletes studied. As such, male gender, older age and Afro-Caribbean origin are all associated with an increased risk [9, 11, 15, 16, 19–21]. For example, the risk of a fatal cardiac event in a black top division US college basketball player during sports was estimated to be as high as 1 : 3000 [17]. Being black is an independent predictor of abnormal ECG findings in healthy subjects but, importantly, black athletes also have an increased risk of SCD compared to white athletes [16, 17, 22], perhaps reflecting a different distribution of underlying disease.

Sudden cardiac death in female athletes

In general, a striking male predominance has been reported for fatal cardiac events during sporting activities [16] (Table 1). However, current data from collegiate athletes refute the notion that SCD is a relevant issue only in male athletes. Indeed, Harmon *et al.* [17] demonstrated a male/female ratio of SCD of only 2.3. A few surveys have focused on sports-related SCD in women. In 2006, Eckart *et al.* [20] reported the findings from a large cohort of female military recruits (median age 19 years) followed for 14 years. The mortality rate was 11.4 per 100 000 person-years; of note, the majority of SCD events occurred in African American recruits (73%) [20]. The lower incidence of SCD in women may be related to the fact that they are less likely to participate in competitive sports compared to their male counterparts. Additionally, male gender itself has been reported to be an independent risk factor for sports-related SCD [not

Table 1 Incidence of SCD or SCA in various cohorts

Study population	Reference	Incidence (person-years)	Male/female ratio
US male Joggers (age 30–64)	Thompson <i>et al.</i> [15]	1 : 7620 (SCD)	–
Competitive Italian athletes (age 12–35)	Corrado <i>et al.</i> [1]	1 : 43 478 (SCD)	9.2
US military (age 18–35)	Eckart <i>et al.</i> [19]	1 : 9000 (SCD)	11.8
US athletes (age 12–35)	Maron <i>et al.</i> [16]	1 : 163 000 (SCD)	8.4
US college athletes (age 17–23)	Harmon <i>et al.</i> [17]	1 : 43 770 (SCD)	2.3
US marathon racers (age 29–55)	Kim <i>et al.</i> [18]	1 : 184 000 (SCA)	5.6

SCA, sudden cardiac arrest; SCD, sudden cardiac death.

only in association with coronary artery disease (CAD) [16], also indicating a variation in the underlying disease panorama.

Is sudden cardiac death only relevant for 'top-level' athletes?

Individuals who participate in competitive sports have a significantly increased risk of SCD compared to those who only take part in low-level leisure-time physical activities [1, 10]. However, the definition of competitive sports is imprecise [16]. According to a definition by the American Heart Association (AHA) [23], 'a competitive athlete is a young (usually <35 years) subject who participates in an organized team or individual sport that requires systematic training and regular competition against others and that places a high premium on athletic excellence and achievement'. Conversely, a leisure athlete is 'an individual, usually middle-aged or elderly (≥ 35 years), who participates in a variety of informal recreational sports, on either a regular or an inconsistent basis, which do not require systematic training or the pursuit of excellence' [24]. These definitions and differentiations are subtle and may mislead athletes as well as physicians. Firstly, the relevance of the problem may be underestimated by excluding all fatal events in the huge subgroup of hobby athletes, who perform sports and therefore have a risk of SCD. Secondly, hobby or non-top-level but nevertheless competitive athletes may be excluded from any cardiac screening. In future, uniform terminology should be used to define an athlete or a competitive athlete.

Importantly, the findings of the recent study by Harmon *et al.* [17] indicated that the risk of SCD increased with the level of sporting competition, being the highest in the top division at college level, compared to lower levels. By contrast, in South Africa, a higher incidence of SCD was found in runners of the half marathon (6.7 per 100 000) compared to runners of the ultramarathon (0 per 100 000) [25], probably reflecting a higher mean risk profile in the former. This is also supported by the progressive increase in SCD, with age, seen in athletes in other studies [21, 26].

Mechanisms and causes of sudden cardiac death in athletes

Independent of the underlying structural or primarily electrical cardiac disorder, the most common cause of SCD during physical activity is abrupt ventricular tachyarrhythmia [1, 6, 10].

As mentioned above, the most important factor is the age of an athlete, as it defines the most probable underlying pathological cardiac precondition and ultimately determines the optimal screening strategy (see below). In 'older' athletes (age ≥ 35 years), atherosclerotic CAD accounts for the vast majority of fatalities [21, 26]. In younger athletes (age <35 years), the most common causes of SCA and SCD during sports are genetic or congenital cardiovascular abnormalities [12, 16, 24].

Genetic or congenital cardiovascular abnormalities in young athletes

The aetiology of SCD in athletes was found to vary greatly in different studies, conducted in different parts of the world. The distribution of confirmed or suspected aetiology, found in studies, is highly dependent on the quality of data, which in turn rely on the quality of the study methodology.

In several North American studies, hypertrophic cardiomyopathy (HCM) has been reported to account for the majority of fatal cases in young athletes [11, 16] (Fig. 1). In Italy, where nationwide preparticipation screening of competitive athletes has been mandated by law since 30 years back, a huge amount of data is available for assessment. These data suggest that arrhythmogenic right ventricular cardiomyopathy/dysplasia (ARVC) accounts for up to one quarter of SCDs in the Veneto region [12, 13]. It is possible that this unexpectedly high rate of ARVC may be related to genetic factors, with this condition being more common in Italy than elsewhere. However, the major reason for the difference in the distribution of aetiologies in Italy compared to the USA may instead be the well-established Italian cardiac screening programme [14]. This programme involves a 12-lead ECG, has high sensitivity to detect HCM and includes an advanced post-mortem examination (histological and genetic) to detect ARVC [14]. In other countries without a mandated screening programme, the distribution of aetiologies is more similar to that in the USA [27]. Interestingly, a recent survey of SCD amongst the population of King County in Washington, USA revealed that in individuals between the ages of 25 and 35 years, only 3% of cases were due to HCM [21].

Other causes of SCD in younger athletes include coronary artery anomaly [13, 16], which may be the third most common cause, and dilated

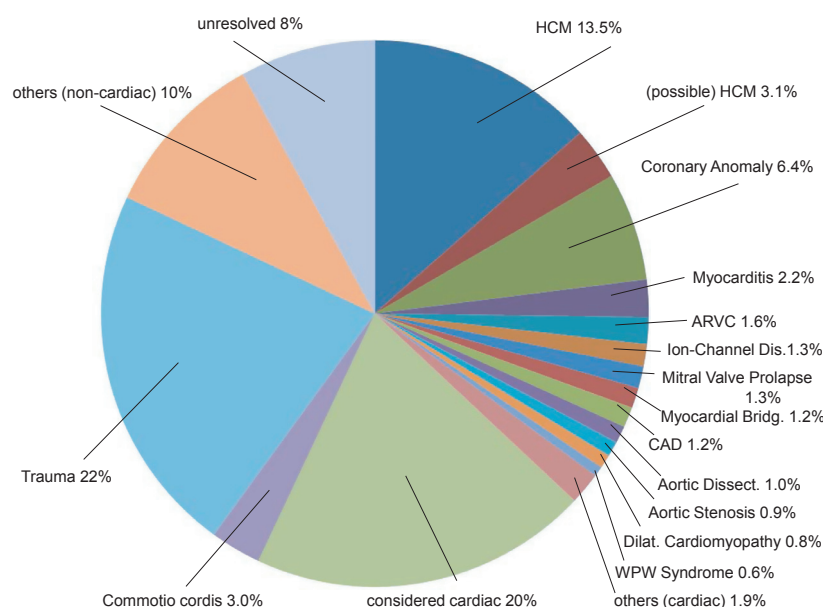


Fig. 1 Causes of sudden cardiovascular death in 1866 young competitive US athletes (adapted from Maron *et al.* [16]). HCM, hypertrophic cardiomyopathy; ARVC, arrhythmogenic right ventricular cardiomyopathy/dysplasia; CAD, coronary artery disease; WPW, Wolff-Parkinson-White.

cardiomyopathy, as well as the more uncommon myocardial bridging [16]. Of note, at least 2–10% of young athletes who die suddenly have no evidence of structural heart disease [16]. In such cases, the cause of a cardiac arrest is probably related to primary electrical heart disease such as inherited cardiac ion channel defects (channelopathies), including long and short QT syndrome, Brugada syndrome and catecholaminergic polymorphic ventricular tachycardia [12, 16, 17, 19]. In addition, SCD may also be related to a nonarrhythmic cause (e.g. spontaneous aortic rupture as a complication of connective tissue diseases such as Marfan's syndrome or associated with a bicuspid aortic valve) [16].

Other reasons for sudden cardiac death in young athletes

Myocarditis has traditionally been considered a major cause of SCD in athletes and is the main reason for the recommendation to abstain from participating in sports during ongoing infection. Today myocarditis is considered to be the cause of death in around 3–8% of athletes [16, 28] but may be much more common in certain subpopulations, such as military personnel [19]. Finally, commotio cordis, caused by a blunt nonpenetrating blow to the precordial area, may provoke spontaneous ventricular fibrillation in structurally and electrically completely normal hearts [29]. In such a situation, only adequate and urgent first responder measures, including the use of an automated

external defibrillator (AED), may save an athlete's life.

Coronary heart disease in older athletes

The risk of sports-related acute cardiovascular events, including SCD, increases exponentially by age of the athlete; this may be related to the development and progression of CAD [21, 26]. More than 80% of SCD events in older athletes are due to underlying CAD [24, 26]. Even in 25- to 35-year-old athletes, 43% of deaths were attributed to underlying CAD in a recent US study [21]. These findings suggest that the age cut-off separating younger from older athletes, traditionally set at age ≥ 35 years, perhaps should be lowered somewhat.

The risk of SCD in an individual with CAD increases transiently with the intensity of exercise [6, 9, 30]. In fact, exercise itself is a trigger, increasing the risk of SCD between 2- and 56-fold [6]. However, regular exercise will decrease the risk of SCD in the long term [7, 31, 32], primarily by positive effects on traditional risk factors, endothelial function, platelet aggregation as well as susceptibility to arrhythmias [33, 34].

Today, a growing number of mainly middle-aged or elderly individuals are engaged in competitive sports activity (for example, marathon running). Marijon *et al.* [26] examined the incidence of sports-related deaths in the general population and reported that

only 6% of fatal events occurred in young competitive athletes, whilst more than 90% were in athletes performing so-called 'recreational sports' at an average age of 46 years. In a Swedish study of 185 athletes >55 years of age, aiming to compete in a 30-km cross-country race, four (2.2%) had severe underlying (silent) cardiovascular disease, including triple-vessel disease and aortic dissection [35].

In the future, an even larger number of older individuals can be expected to take part in different sporting competitions, following national initiatives to encourage individuals to be more physically active. Importantly, participants may be older and less fit, compared to a few years ago. For example, the Swedish study of cross-country racers showed that the runners in all age groups had a progressively longer average running time each year of the study period from 1993 to 2007 [36]. In conclusion, it must be emphasized that the area of SCD in sports has become an important issue of public health in general, not only amongst young, competitive athletes.

How can sudden cardiac death be prevented in athletes?

The rationale for preventing SCD in athletes is mainly based on the fact that certain individuals have an underlying cardiovascular disorder, often unknown, that carries an increased risk of a fatal event [37]. The general assumption, based on the prevalence of such disorders, is that up to 1/300 athletes may have such underlying disease [38]. Cardiovascular screening of athletes is based on identifying individuals at increased risk, somewhat in parallel to identifying those with hypertension, to prevent future cardiovascular events. Recommendations for cardiovascular screening of young athletes [37, 39] have been accompanied by recommendations also for screening/assessment of older or even recreational athletes, willing to be engaged in leisure-time or competitive sports [40].

As cardiovascular screening is never foolproof and thus cannot identify 100% of athletes at risk, such a programme should be complemented by a second line of defence, that is, adequate preparedness at the sporting arena/facility in case of a potentially fatal event.

Cardiac adaptation to training

Before discussing cardiovascular screening in detail, the normal physiological cardiac adaptation

to training should be briefly considered. Unfortunately, such changes are sometimes difficult to differentiate from pathological changes. The heart is a muscle that adapts to training, similar to other muscles in the body, through hypertrophy and maximizing function. The cardiovascular adaptations to training are usually referred to as 'athlete's heart' and include morphological (eccentric muscle hypertrophy and chamber dilatation), electrophysiological (increased vagal tone and bradycardia) and functional changes (augmented stroke volume and increased diastolic function) [41]. These changes may vary according to many different factors, including the type of sport and duration and level of participation, as well as the age, sex and ethnicity of the athlete [42]. In recent years, it has been increasingly considered that sport *per se* may, in extreme cases, even lead to the development of cardiac abnormality (possibly in genetically susceptible individuals). This has been suggested particularly in cyclists, who in some cases have developed right ventricular changes similar to those of ARVC. This condition has been termed 'exercise-induced ARVC' [43, 44]. Although, the debate is still ongoing, there is little doubt that sport may have an 'upper limit' of health benefit and that at least extreme elite sports have the potential to aggravate underlying cardiovascular abnormalities, including HCM and/or arrhythmia risk.

Cardiovascular screening

Cardiovascular screening of young athletes

Cardiovascular screening of young competitive athletes is advocated by both the AHA [39] and the European Society of Cardiology (ESC) [37]. Historically, cardiac screening was part of preparticipation screening already in the 1950s. Then, for a number of years, screening was not considered a high priority; ironically, as at this time, increasingly more was being understood about SCD in athletes and the normal physiological adaptation to exercise. However, today almost all professional organizations advocate cardiac screening, which according to the AHA is 'justifiable, necessary, and compelling on the basis of ethical, legal and medical grounds. Indeed, preparticipation screening is viewed as an important public health initiative' [39]. Sporting bodies have also adopted these recommendations; the Fédération Internationale de Football Association (FIFA) and the Union of European Football Associations (UEFA) have in recent years made cardiac screening mandatory

before any competition, and the International Olympic Committee recommends it as 'best standard' practice [45]. In fact, this implementation of cardiac screening in elite athletes is parallel to the regular screening of others with physically demanding jobs, including firemen, divers, pilots and military personnel.

The most important question now is not whether to screen, but rather how to screen. In particular, whether or not to include the ECG as part of basic screening has been debated in recent years. The ESC recommends the inclusion of the resting ECG, in addition to family/personal history and physical examination [37] (Fig. 2), whereas the AHA advocates only personal/family history and physical examination [39]. Recently, this European/American debate has been somewhat less intense, as increasingly more international bodies, as well as US physicians and professional leagues, do in fact include the ECG as part of their screening routine. However, the best standard for cardiac screening remains to be fully determined, and the current recommendations may be updated as more evidence emerges for the usefulness of different imaging techniques, cardiac biomarkers and/or genetic markers. What is generally agreed upon is that if the basic screening is 'normal', the athlete is eligible to compete, and if the initial screening is abnormal, the athlete may need to undergo further examination to confirm and/or exclude an underlying cardiovascular disorder [37, 39].

The only prospective large-scale study to investigate the efficacy of cardiac screening of young athletes was conducted in Italy [14]. The Italian experience of mandatory screening of all competitive athletes from 13 or 14 years of age, through

personal and family history, physical examination and resting ECG, was evaluated. The implementation of this screening model from the end of the 1970s until the early 2000s was associated with a reduction in the incidence of SCD in athletes from 4.0 to 0.4 per 100 000, whilst the incidence in the general population remained unchanged at 1.0 per 100 000 [14]. Of note, the decrease in events was the result of fewer deaths due to HCM, the most common cause of SCD, and screening proved most effective at identifying individuals with this condition.

It is clear that more prospective studies showing the efficacy of cardiac screening are required, as this Italian study was nonrandomized. Furthermore, some studies have relied on media information for collecting SCD data in athletes, which has been shown to miss at least 50% of cases [17]. High-quality studies, with uniform methodology and definitions, need to be performed. For this purpose, the establishment of prospective uniform registries of SCA/SCD in athletes is the highest priority.

Electrocardiogram interpretation in athletes

The resting ECG has been shown to be crucial for cardiac screening, having a high sensitivity for detecting underlying diseases associated with SCA/SCD in athletes. For example, adult patients with HCM have typical ECG changes in 90–95% of cases [46]; in addition, ARVC frequently presents with ECG abnormalities [38]. Electrophysiological abnormalities such as long and short QT syndrome, Brugada syndrome, high-grade atrio-ventricular (AV) block and the delta-wave characteristic of Wolff–Parkinson–White (syndrome) are usually detected by ECG [38]. Whilst some underlying disorders, such as early ARVC and coronary artery anomaly, may not show any ECG changes, the majority of cases of SCD will show some ECG abnormalities [47].

However, as mentioned above, normal cardiac adaptation to training (athlete's heart) may also include ECG changes, such as sinus bradycardia, sinus arrhythmia, AV block I (and possibly IIa), as well as repolarization changes and isolated voltage criteria of left ventricular hypertrophy, which all may be considered abnormal in a nonathlete population [38]. This has led to the assumption that the ECG in athletes is 'often or always abnormal'. Indeed, the ECG of a healthy athlete may

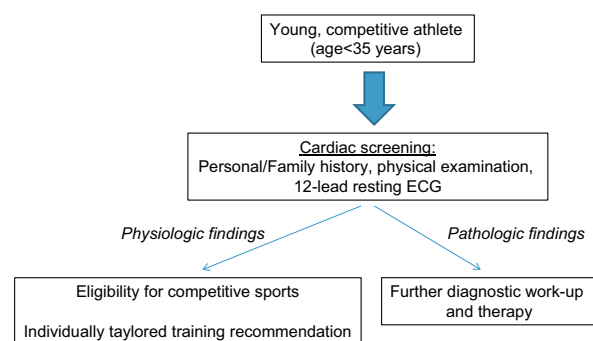


Fig. 2 Screening recommendations for young competitive athletes (<30–35 years).

often be considered 'abnormal' by the automated ECG interpretation system, causing the problem of false-positive ECGs and the potential cost for further investigation. However, Pelliccia *et al.* [48] demonstrated that only around 10% of >1000 athletes had abnormal ECGs, and only 4.8% of these being changes not associated with normal physiological adaptation. It is very important to achieve a high specificity, in addition to the known high sensitivity, for the ECG to become an accepted part of the routine screening of athletes.

Of importance, as outlined in the recent ESC recommendations for interpretation of the ECG in athletes [38], the vast majority of ECG changes (80–90%) are considered to be benign, reflecting cardiac adaptation to training, and do not require follow-up. Only a minority (5–10%) of ECG changes seen in athletes are considered to be a possible sign of underlying disease [38, 49]. Of even more importance, incorporation of new data on ethnicity and gender has improved the ECG interpretation, by adjusting the borders of 'normality' to some degree [49]. The newest European recommendations on ECG screening were found to have increased specificity without lowering the high sensitivity [50].

A summit of international experts was convened in Seattle in February 2012 with the main goal of producing clear, clinically relevant and easily accessible recommendations for interpretation of the ECG in athletes. The 'Seattle criteria' have been published [49] and are also available as an online educational tool (<http://learning.bmj.com/ECG-athlete>). The ECG changes in athletes are categorized as 'normal' – no need for further check-up or 'abnormal' – needing further investigation' (Fig. 3). These recommendations have the potential to further reduce the false-positive rate of ECG findings.

To summarize, screening including an ECG recording has a high sensitivity for underlying disease, but the specificity needs to be improved, whereas screening of athletes without the use of the ECG has a very low sensitivity [51]. However, the most important, and somewhat neglected, argument for including the ECG may be the even lower specificity of screening in its absence. By using only family history and symptoms, the screening physician will uncover a multitude of symptoms, often diffuse and difficult to explain [52]. The most common first line of follow-up would then be the ECG, so that it

would be necessary in many cases anyway. Cost-effectiveness studies are always difficult to interpret, but most seem to favour the inclusion of the ECG in screening [53].

Screening of older/master athletes

Because of the known health benefits of regular exercise, not least to prevent cardiovascular morbidity and mortality [3], promotion of increased physical activity, including sports, is a priority for all ages. To achieve the maximum benefit of activity, whilst minimizing the risks, exercise regimens should be individually tailored [40, 54].

As described above, we know that CAD is the most common cause of SCD in athletes >30 years of age, that the risks of CAD and SCD increase progressively with age and that the risk of SCD increases with the intensity of the sporting activity [21, 26]. As a number of individuals prepared to participate also in challenging competitions such as marathon running or cross-country racing, also at advanced age, have (silent) underlying CAD [35], the most important question is, can we identify these individuals?

Cardiac screening of older/elderly athletes is different from that of younger athletes. For example, the resting ECG is of limited value to detect silent CAD, whereas evaluation of traditional risk factors and family history of CAD are essential [54]. Both US (AHA) [55] and European (European Association for Cardiovascular Prevention and Rehabilitation; EACPR) [40] guidelines advocate at least some degree of cardiac screening of elderly athletes. The aim of such screening is to identify those individuals at highest risk, who need further evaluation, using a practical method of self-assessment. The AHA recommends maximal exercise testing of all competitive athletes above the age of 40 (in men) or 50 years (in women) with one additional risk factor (according to blood tests or family history) [55]. The EACPR recommends an individual approach, in which the level of testing required depends on the individual risk profile (family history, symptoms, risk factors and age, as well as the habitual level of physical activity) and the type of intended activity [40]. For example, a low-risk individual who runs regularly without symptoms does not need to be screened in order to take up golf, whereas an overweight, diabetic subject should be screened before taking part in a half marathon race [40].

<u>Normal ECG findings in athletes</u> - common <u>training-related</u> ECG alterations - <u>physiological adaptations</u> to exercise Do not require further evaluation	<u>Abnormal ECG findings in athletes</u> - <u>unrelated</u> to regular exercise Require further diagnostic evaluation
- Sinus bradycardia (≥ 30 bpm) - Sinus arrhythmia - Ectopic atrial/junctional escape rhythm - I° or II°a/Wenckebach AV block - Incomplete right bundle branch block - Isolated QRS voltage criteria - Early repolarisation	- T wave inversion (excl. III, aVR, V1) - ST segment depression (≥ 0.5 mm) - Q waves (excl. III and aVR) - Complete left bundle branch block - QRS duration ≥ 140 ms - Left axis deviation - Left atrial enlargement - Right ventricular hypertrophy pattern - Ventricular arrhythmias - Wolff-Parkinson-White pattern - Long- or short-QT interval - Brugada-like ECG pattern - Profound sinus bradycardia < 30 bpm - Atrial tachyarrhythmias - Ventricular arrhythmias

Fig. 3 Physiological (normal) and abnormal ECG findings in athletes (adapted from Drezner *et al.* [49]). AV, atrioventricular; bpm, beats per minute; ECG, electrocardiogram.

It is important to note that the efficacy of these recommended screening methods has not been proven. A small study of middle-aged, first-time cross-country runners recently showed that a similar evaluation, to the one recommended by the EACPR, could identify individuals in need of further testing [56].

Arena safety

Cardiac screening, however thorough, will never be able to detect all individuals at risk. Screening will have the highest yield for macroscopic structural disease, such as cardiomyopathy, valve disease or electrophysiological disorders, evident on the ECG [37]. However, early structural disease such as early ARVC may be very difficult to identify on screening [38]. Furthermore, the percentage of SCDs in athletes with supposedly 'normal hearts', subsequently needing genetic testing, to establish the cause, is of course also very difficult to detect on ECG, although they may have symptoms and/or family history of SCA/SCD.

In the case of a serious adverse event, such as SCA, the application of 'the chain of survival' [i.e. immediate cardiopulmonary resuscitation, fast defibrillation (if needed) and advanced cardiac life support (ACLS)] will increase considerably the chance of survival [57]. In out-of-hospital settings, the highest survival rates are seen in locations within close proximity to an AED, such as airports [58]. In the US high school setting, the rate of survival

following SCA in relation to sports rose as the result of increasing use of AEDs (during training and competitions) [59].

The incidence of SCA in sports arenas is estimated to be around one in 500 000 spectators [60, 61] or even higher [62]. The majority of cases are men with ventricular fibrillation as the initial rhythm [61, 62], and an organized response system, including prompt defibrillation, may achieve a survival rate exceeding that usually obtained by public healthcare systems [61, 62].

The importance of emergency preparedness at sporting arenas has gained an increasing awareness in recent years. The Arena study [60] showed that emergency preparedness was inadequate in a large proportion of 190 major football stadiums in Europe. For example, 35% did not have a written medical action plan (which should include information regarding 'who does what' in the case of emergency, the responsible doctor, available equipment, and personnel as well as the proper training of these, and how contact with the local ACLS system is established), and 20% of the arenas with a transport time of more than 10 min to the nearest hospital did not have an AED [60]. Improvements are also needed in the USA [63].

Recommendations for arena emergency preparedness

As a result of the above studies, recommendations for proper emergency preparedness have been

published recently by the EACPR for implementation in sports arenas [64] and in the USA for secondary schools [65]. Of note, these recommendations have also begun to be applied by sporting bodies. For example, the UEFA has mandated a high higher standard at all UEFA events, including listings of required equipment, personnel and ambulances at both training sessions and matches. This information should also be made available to the visiting team. In addition, UEFA has introduced a training course for international doctors from all member countries in acute emergency care on the playing field. Similarly active in this area, the FIFA has supplied all football associations with a fully equipped 'medical emergency bag', which includes an AED, and produced guidelines for preventing SCD in football (FIFA 11 steps to prevent SCD) [66].

Many athletes of different ages and risk levels also take part in sporting activity at exercise facilities such as gyms. Little is known about the current situation regarding emergency preparedness at these facilities, but an ongoing study, organized by the EACPR, will target this issue.

Summary

Sudden death of athletes is usually cardiac in origin. In fact, in most cases of SCD, there is an underlying, often clinically silent, cardiovascular abnormality. The incidence of SCD in young athletes is low, but attracts considerable media attention. Furthermore, the incidence of SCD in athletes overall may be expected to rise as increasingly more, as well as older and possibly less fit, individuals are being encouraged to take part in organized sporting activity. Whilst exercise and sports have scientifically proven health benefits, individuals at high risk may have to tailor their exercise to optimizing their risk/benefit ratio.

For athletes of all ages, cardiac screening may identify individuals at risk and has the potential to decrease the risk of SCD in relation to sports, as indicated by the Italian data [14]. Importantly, the content of cardiac screening will have to be different in younger and older athletes, due to the different underlying disorders causing SCD in the different age groups. In addition, cardiac screening must be complemented by appropriate safety measures at sporting facilities/arenas, to increase the chance of survival in case of a potentially fatal event.

More information about the incidence/aetiology of SCD as well as the appropriate cardiac screening programmes in different subgroups of athletes is needed. The availability of such data in the future is dependent on the quality of study methodologies. The establishment of uniform SCD registries in athletes is a future vision that would make it possible to investigate regional differences in SCD aetiology, survival and the efficacy of different screening programmes. An initial step towards this vision is the ongoing project from the section of Sports Cardiology of the EACPR to produce a framework for such a registry.

Conflict of interest statement

No conflict of interest was declared.

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